Feasibility-First Humanoid Motion Tracking: Screening Reference–Physics Misalignment and a Matched Test of Admission and Allocation

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Abstract— **Persistent tracking failure need not identify useful practice: a retargeted reference may demand support unavailable under the declared robot and scene. We study this reference–physics misalignment through model-relative screening and matched training experiments. In a three-seed G1 campaign, exposure peaks at 87–89%, an unsupported kneel-and-crawl reference repeatedly attracts practice, and uniform sampling yields higher held-out survival in each seed. A contact-capacity screen flags 2,442/10,705 AMASS-derived references in one pipeline and 7/4,950 in a separate production pairing; two implementations agree on 39/40 enriched-panel decisions. Feasibility features improve cross-policy difficulty correlation from 0.567 to 0.609 on 100 held-out clips. Exact temporal support then separates two interventions at 49.2 million transitions per policy. The completed three-seed allocation comparison is inconclusive: feasible-hard relative-progress minus uniform TrackingScore is** -0.0151 **with paired 95% seed-t interval** $[-0.0944, +0.0642]$**, and its broad non-regression guard fails. A separate five-seed admission on/off comparison under conditional-failure allocation is** ***inconclusive*****: feasible-hard on-minus-off is** $+0.0000$**, interval** $[-0.0196, +0.0196]$**. These tests distinguish model-relative qualification, changed exposure and measured control utility; none alone establishes physical feasibility or general transfer.**

I. Introduction

Large retargeted motion banks have made generalist humanoid tracking a practical training recipe: map human motion to a robot, train a policy over the resulting references in massively parallel simulation, and spend additional updates on motions the policy currently fails. Systems such as PHC, MaskedMimic, H2O, ExBody, BeyondMimic, and SONIC demonstrate the scale of this recipe [1], [2], [3], [4], [5], [6]. Failure, completion, or recent learning progress then appears to identify where the next policy update is most useful [5], [7], [8].

That interpretation assumes every reference defines a solvable control problem. A retargeted trajectory may instead have no admissible contact capable of supplying its demanded base wrench, or may require joint effort beyond the modeled actuator range. We call this *reference–physics misalignment* (RPM). Under RPM, failures can reflect reference inadmissibility as well as limited experience or control capability. Our implemented screen addresses the first distinction. A coverage floor cannot create a behavior absent from the bank, and a progress estimate does not identify intrinsic difficulty or forgetting. The experimental question is therefore: given fixed reference support and a training budget, does learner-dependent allocation produce better tracking on the same held-out objective?

We observed the consequence in a 100-motion Unitree G1 campaign. A BeyondMimic-style sampler concentrated on the same kneel-and-crawl reference in all three seeds. Campaign peak top-1 mass was 0.884, 0.870, and 0.893; another clip owns the maximum in two seeds, but the same kneel-and-crawl reference is the recurrent attractor. It passes ordinary kinematic checks. Yet during its 0.75–1.75 s descent, the nearest collision geometry remains 7–10 cm above the floor while a median 329 N of support demand—approximately the 327 N model weight—has no modeled source.

CLIMB makes reference admission and practice allocation separately testable (Fig. 1). The `refeas` screen evaluates final robot-space references under a declared model; temporal units enforce support throughout each trial. A bounded allocator may use policy outcomes to change exposure within that support. Whether this feedback improves control is an empirical question, not a property of the interface.

Our contributions are:

1) an RPM diagnosis and model-relative contact-capacity screen, evaluated at bank scale and across implementations, with feasibility features that transfer difficulty information between policies sharing a learner;
2) an exact temporal-support interface and a matched four-arm, three-seed allocation study with a complete, inconclusive policy result and explicit exposure measurements;
3) a separately frozen five-pair admission on/off test under conditional-failure allocation, reporting its effect and rejected-support exposure without conflating admission with the allocation comparison.

We also report a bounded repair alternative: 22/26 screened candidates qualify, yet a one-policy paired deployment comparison does not establish an aggregate tracking gain. The separate admission comparison changes the admission mask under a common allocator; it does not test an admission-by-relative-progress interaction.

II. Related Work

A. Motion filtering and dynamic retargeting

Policy-based curation already removes poorly tracked references. H2O retains AMASS motions that a privileged imitator can follow, and ExBody2 uses an initial policy's sequence errors to select a feasible and diverse subset [3], [9]. These scores combine reference quality with policy capability. KungfuBot instead uses a human-space center-of-mass/center-of-pressure heuristic before robot retargeting [10]. LIMMT combines target-robot motion heuristics, diversity, and complexity, with physical-score weights calibrated through repeated policy training [11]. PHUMA filters source artifacts and retargets with joint-limit, ground-contact, and anti-skating losses [12]. CLIMB builds on this line but changes the training interface: the final robot-space screen determines the support on which an adaptive allocator is permitted to act.

Optimization-based retargeters address a complementary problem. Kinodynamic Motion Retargeting uses rigid-body and contact constraints to construct viable locomotion, while Direct Dynamic Retargeting uses simulator-in-the-loop optimization without an infeasible geometric intermediate [13], [14]. AMO and SPIDER likewise generate viable references through trajectory optimization or physics-based sampling with contact guidance [15], [16]. GMR further shows that retargeting artifacts affect downstream tracking robustness [17]. An earlier analytic study estimates velocity, acceleration, and torque requirements for 40 upper-body motions, but does not test whole-body environmental support [18]. Repair can therefore be preferable to exclusion. Our narrower question is diagnostic: after retargeting, can the demanded robot-space wrench be supplied by admissible contacts within modeled actuator limits?

B. Adaptive allocation and evaluation

Prioritized experience and level replay formalize the benefit and bias of nonuniform training distributions [19], [20]. In humanoid tracking, BeyondMimic uses a failure-EMA bin sampler, while GMT and EGM reweight motions or segments using tracking outcomes [5], [7], [8]. Those systems combine allocation with curation, staging, clipping, or architectural changes. Empirical failure, completion, and tracking error identify hard examples, but do not by themselves determine whether a reference is dynamically admissible. CLIMB inserts that test before outcome-based allocation; its matched comparison then changes only allocation over a shared set of legal trials. Learning-progress curricula already include ALP-GMM and portable implementations such as Syllabus [21], [22]. The contribution is the temporal-support contract and controlled test of policy utility, not normalization of a progress score alone.

Robot-learning comparisons are sensitive to seed count, aggregation, and interval construction [23]. Tracking adds a conditioning problem: kinematic error among survivors can improve when a method terminates earlier. HumanTracker similarly finds that kinematic metrics can miss support and contact failures [24]. We therefore use a liveness-weighted primary score and report common-survivor quality only beside survival.

III. Screening and Exact Temporal Support

A. Model-relative feasibility

Let a retargeted reference provide generalized position and velocity $(q_t, \dot{q}_t)$ for the G1 at frame t. We apply a centered five-frame moving average to velocity, differentiate at the reference frame rate, and smooth acceleration with the same window. Feasibility is defined relative to a robot model, scene, collision geometry, friction, actuator ranges, and reference. The present instantiation uses the G1 and a flat plane; electrical, thermal, compliance, latency, and structural limits remain outside this model-relative label.

We use the compiled MuJoCo model underlying the tracking environment. Contact-free inverse dynamics returns the generalized force required for the prescribed acceleration. The six free-joint coordinates define the base wrench $\mathbf{W}_t$ that the environment must supply; the remaining coordinates define contact-free joint torque $\boldsymbol{\tau}_{\text{free},t}$. A collision geometry becomes a candidate contact when its exact distance to the plane is at most $g = 0.06\,\text{m}$.

The 6 cm band is a declared tolerance. At 3 cm a known feasible control is flagged on 42% of frames because the bank carries an approximately 3 cm ground-alignment offset. At 10 cm the attractor's 7–10 cm floating descent is incorrectly granted a contact candidate. The chosen setting lies between these observed failure modes; it is not a universal constant.

B. `refeas`: contact-capacity screen

For each candidate point we construct a four-edge pyramidal friction cone with coefficient 0.6 in the primary model; a simulator-matched view makes non-foot contacts frictionless. Mapping nonnegative generator magnitudes f_t through the contact Jacobian yields both base-wrench matrix $\mathbf{A}_t\mathbf{f}_t$ and joint-torque contribution $\mathbf{J}_t\mathbf{f}_t$. A weighted nonnegative least-squares problem first exposes the unconstrained residual, with angular-wrench rows weighted by two. We then solve a linear program minimizing ℓ_1 wrench slack subject to

$$-\boldsymbol{\tau}_{\max} \le \boldsymbol{\tau}_{\text{free},t} - \mathbf{J}_t\mathbf{f}_t \le \boldsymbol{\tau}_{\max}, \qquad \mathbf{f}_t \ge 0. \tag{1}$$

The translational component of the resulting weighted residual defines unsupported force. With no candidate contact, the residual is the contact-free demand.

We report the fraction of frames with no contact candidate (`airborne_frac`), the fraction whose torque-limited unsupported force exceeds half the robot weight (`infeasible_frac`), normalized unsupported-force integral, and joint-torque infeasibility. A clip crosses the primary rule only when `infeasible_frac`> 0.10 (strict inequality). Ballistic flight is not flagged merely for being airborne: near-gravity acceleration has little external support demand. Conversely, nonballistic hovering or descent without support is flagged.

The screen takes approximately one CPU-second per clip in the primary implementation. An independent implementation processes the production bank at 0.145 CPU-s per clip (0.84 ms/frame). The screen therefore runs as an offline bank-ingestion stage rather than in the policy control loop.

C. Repair as a routing alternative

DFRP applies a bounded contact projection to qualified floating-reference cases. It lowers and smooths root clearance, then adjusts supporting-leg joints with damped contact inverse kinematics and joint-limit projection. The changed trajectory is re-screened and must satisfy residual infeasibility at most 5%, root displacement at most 8 cm, valid joint limits, contact-IK residual at most 10 mm, and at least one legal 50-step start. Missing-scene cases bypass projection. This is an implemented routing option, not a guarantee of usable control: repair changes the tracking target and needs its own paired policy evaluation.

D. Exact-support gated allocation

Clip classification is too coarse because one reference can contain both supportable and unsupported intervals. We reduce per-frame screens to ordered feasible runs, retain only 50-step windows wholly inside a run, and bind each motion and sidecar by SHA-256. Across 800 training clips, 1,841 source runs yield 1,184 admissible units and 368,951 legal starts after 657 short runs are discarded. A unit is an attribution stratum; the deployment prior is uniform over legal starts, not units or clips.

For an admitted interval $\mathcal{A}_u$, define legal starts by every reference frame accessed during horizon H and lookahead configuration $\mathcal{L}$:

$$\mathcal{S}_u = \{s : s + \mathcal{K}(H, \mathcal{L}) \subseteq \mathcal{A}_u\}. \tag{2}$$

The current interface includes the terminal observation and uses current-frame reference observations. “Exact” denotes membership relative to this screen and access contract; it is not a certificate of physical feasibility.

Let b_u be the normalized legal-start prior over admitted units and $g_u = |s_u(k) - s_u(k-10)|$ the nonnegative change of Beta-smoothed conditional success at sampler clock k. Uniform U uses b. Absolute-progress A retains the earlier fixed additive floor. Relative-progress R, before active caps and for complete history with $\mu = \sum_u b_u g_u > 0$, is exactly

$$p_u = 0.8 b_u + 0.2 \frac{b_u g_u}{\mu}. \tag{3}$$

The implemented prior/cap fallback covers incomplete history or zero μ. Common positive rescaling of g leaves Eq. (3) unchanged, and its pre-cap total variation from b is at most 0.2. These are exposure properties, not guarantees of useful practice. D uses conditional failure under the same support; its startup and floor/cap composition differ from R, so R−D compares allocator designs rather than isolating ranking alone.

Intervals do not all receive the same remedy. Admissible runs and DFRP-qualified repairs enter the exact support; missing-context and residual failures do not. This routing preserves feasible subsequences rather than treating every screen flag as a whole-clip deletion instruction. The four-arm comparison uses the original admitted bank; it substitutes no repaired targets.

The runtime samples only legal starts, emits explicit truncation at a segment boundary, never wraps into a rejected frame, and serializes its deterministic generator and sufficient statistics. Per-unit and per-clip caps bound concentration. Missing sidecars, changed motion hashes, invalid starts, invalid frames, or censored resets fail the contract.

The paired evaluator freezes the motion, start, replicate, environment-noise seed, and dynamics-noise seed across policies. It assigns the reference before the first observation, disables auto-reset until terminal channels are captured, rejects invalid offsets, and records checkpoint, task, condition, reference, evaluator, and output hashes. The 100-clip panel is name- and hash-disjoint from training and contains 2,800 paired conditions.

IV. Experimental Design

A. E1: RPM attractor diagnosis

We analyze the previously sealed 100-motion campaign: 4,096 environments, 4,000 PPO iterations, three seeds per arm, and 100 disjoint evaluation motions with eight episodes per clip. It includes failure-adaptive, clip-uniform, and normalized grounded samplers. We report exposure, attractor identity, held-out survival, and the attractor’s modeled contact-capacity residual. This campaign establishes the motivating failure case; the later four-arm study separately compares exact-support allocators.

B. E2: bank scale, agreement, and transfer

The primary corpus contains 10,705 AMASS-derived references retargeted through `whole_body_tracking` to G1. We report raw counts under the fixed rule, category/source breakdowns, and contamination in the historical evaluator. A separate experiment covers all 4,950 references in the filtered BONES-SEED production pairing. We never pool their prevalence. A deterministic 40-clip panel, 20 from each bank and enriched for flags, passes through both implementations to measure score and decision agreement; it is not a prevalence sample. Separately, a ridge model fits 11 intrinsic reference features to per-clip difficulty under one curriculum and ranks difficulty under another on the same 100 held-out clips. We compare intrinsic features with and without the three `refeas` features against 200 random three-feature additions; this is cross-policy transfer, not cross-architecture transfer.

C. E3: DFRP repair qualification

We freeze a source- and severity-stratified CPU panel before repair: 26 strict-flagged candidates span three root-displacement bands and two initial-infeasibility bands, and four source-matched feasible clips serve as no-op controls. The primary outcome is exact-ready flagged count under the 5% residual, 8 cm root, joint-limit, 10 mm contact-IK, source-hash, and legal-start gates. The gate requires at least 20/26 candidates, 4/4 byte-identical ready controls, zero integrity or qualification failures among admitted clips, and a reproducible manifest payload. Runtime and root, joint, body, velocity, and acceleration deviations are secondary diagnostics.

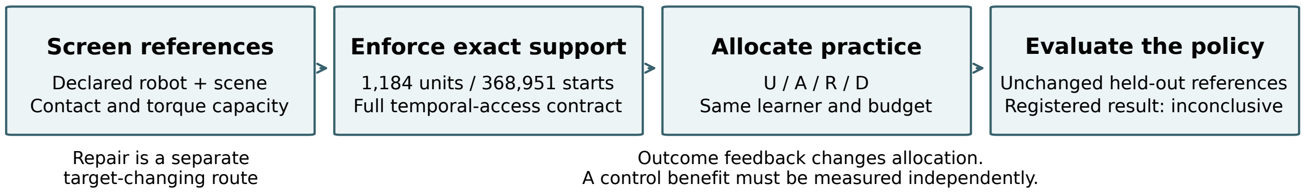

Fig. 1. **Admission, allocation and policy utility are separate tests.** The screen supplies model-relative support; the temporal interface enforces legal reference access; independent evaluation measures the resulting policy. The tested outcome-to-allocation feedback changed exposure but did not establish the registered tracking benefit. Neither screen qualification nor exact support implies hardware feasibility. Repair is a separate, target-changing route.

D. Four-arm allocation on unchanged exact support

An earlier two-arm E4 protocol selected absolute-progress settings using allocation-only calibration. Its long-run manipulation check subsequently failed, leaving policy endpoints closed. A separate prospective confirmation retained A as a declared comparator and added relative progress R and conditional failure D alongside uniform U.

The completed study uses seeds 21, 22 and 23, 512 environments, 4,000 PPO iterations and 24 rollout steps: 49,152,000 simulator transitions per policy. Support, learner, rewards and reference targets are common. All twelve training gates passed before any held-out cell opened. Checkpoints 1000, 2000, 3000 and 3999 give 48 evaluation cells on 100 unchanged held-out clips, including 25 feasible-hard clips selected from reference features. Each cell contains 2,800 conditions, evaluated over uninterrupted windows of up to three seconds.

The actor and critic are observation-normalized ELU multilayer perceptrons with hidden widths 512/256/128, taking 160 and 286 observation coordinates, respectively; the actor outputs 29 joint-position commands. Physics runs at 200 Hz with 50 Hz control. PPO uses Adam (initial learning rate 10^{-3}, adaptive KL target 0.01), five epochs, four minibatches, clipping 0.2, discount 0.99, GAE coefficient 0.95 and entropy coefficient 0.005. These settings are shared across arms.

The primary score is liveness-weighted reference tracking:

$$h \exp\left(-\frac{\text{MPKPE}}{0.30\,\text{m}} - \frac{\text{anchor error}}{0.40\,\text{rad}}\right), \tag{4}$$

where h is survived-horizon fraction, MPKPE is root-relative body-position error, and anchor error is orientation error. We average conditions within clip, clips within panel and paired differences across training seeds. Independent replication is the trained seed pair, not the number of conditions.

The frozen primary decision requires final hard R−U mean at least +0.02, a positive lower two-sided 95% paired seed t bound (df=2), and all-panel lower bound above −0.01. A pass would establish a positive effect with a target-reaching point estimate, not a true effect of at least +0.02. Paired hierarchical bootstrap intervals are supplementary. Learning curves, normalized area under the observed curve (AULC), and attainment of the paired U final score are exploratory; non-attainment is retained. No seeds are appended to the completed decision.

E. H1: matched admission under conditional failure

The separately frozen H1 study compares D with admission on/off over a common candidate partition: 1,184 admitted and 465 rejected units, with 368,951 of 417,072 legal starts admitted. On renormalizes the legal-start prior over the admitted subset; off retains all candidate units. Both retain non-wrapping 50-step trials, the common learner, budget and unit/clip caps. Five fresh paired seeds (1041–1045) yield ten training runs; all training gates precede forty evaluations on the same 100-clip panel and checkpoint grid. This panel is reused from the allocation study. The final hard on-minus-off two-sided seed-t interval (df=4) determines positive (lower > 0), negative (upper < 0), or inconclusive. The +0.02 line is descriptive; all-panel and bootstrap intervals carry no additional decision rule. The mechanism endpoint attributes positive excess above the prior to rejected units using equal-weight saved checkpoints 1000–3900 by 100 and 3999; it also records top-1 identity.

V. Results

A. E1: Curriculum collapse under RPM

Adaptive peak top-1 mass is 0.884/0.870/0.893 across campaign seeds, and the same BMLmovi kneel-and-crawl clip repeatedly becomes dominant in all three. The maximum belongs to another clip in two seeds. Uniform sampling achieves higher held-out endpoint survival in every paired seed: differences (uniform minus adaptive) are +0.0300, +0.0275, and +0.0300. With only three seeds, the sign pattern is the useful description; the minimum attainable paired permutation p-value is 0.125.

The reference has no joint-limit violation and peak joint speed is at most 5.6 rad/s. The dynamic screen instead localizes the failure. From 0.75–1.75 s, 86% of frames have no candidate contact and median unsupported force is 329 N; the model weighs 327 N. The 1.75–7.25 s kneeling interval is supportable through non-foot contacts, and the 8.0–8.5 s rise again becomes unsupported. Every trained policy has zero survival from frame zero, while the tested late 8 s offset survives.

B. E2: Bank-scale screening and difficulty transfer

Under the strict rule, 2,442 of 10,705 primary-pipeline clips are flagged (22.8%). Category rates range from 12.9% for quiet motion to 58.6% for dynamic motion, and source rates from 0.1% for GRAB to 100% for CNRS. Severity-stratified inspection shows why labels are insufficient: five inspected CNRS cases are ordinary fast walks with extended floating intervals, while the Transitions sample mixes ingest defects with acrobatic content.

Only 7 of 4,950 production clips (0.14%) cross the same nominal rule, while 111 (2.24%) are airborne on more than 10% of frames. Five of seven flags are jumps, four naming a 50 cm box absent from the declared flat scene. Their verdict is *unsupported on this plane*, routing them toward missing terrain or exclusion rather than geometric root projection.

Across the stratified 40-clip panel, `infeasible_frac` rank correlation is 0.9836, `airborne_frac` correlation is 0.9974, and strict decisions agree for 39/40 clips (Cohen's $\kappa = 0.9485$). The disagreement is `burpee_002__A362_M` (0.019 versus 0.136). Figure 2 keeps the full-bank denominators separate and uses only this same-input panel to check implementation agreement; it does not establish corpus or retargeter equivalence.

The difficulty decomposition is also visible across policies trained with different curriculum allocations. On the same 100 held-out clips, an 11-feature intrinsic atlas fit to adaptive-policy difficulty ranks uniform-policy difficulty at $\rho = 0.567$. Adding `infeasible_frac`, `airborne_frac`, and normalized unsupported impulse raises the rank correlation to 0.609, above 198 of 200 random three-feature additions (one-sided $p = 0.010$). All six directed policy pairs move in the same direction and four have $p < 0.05$; the two pairs targeting the grounded policy do not. This is transfer across policies sharing an architecture, not evidence of cross-architecture or cross-robot transfer.

C. E3: Contact-manifold repair via DFRP

DFRP qualifies 22/26 flagged candidates (84.6%) and all four feasible controls pass as byte-identical no-ops. The resulting 26-clip curated view contains 36 admissible units and 10,561 legal 50-step starts. Median and 95th-percentile CPU runtime are 2.57 and 6.29 s per clip. Two candidates remain above the 5% residual-infeasibility ceiling, and two more reach that ceiling but violate the 10 mm contact-IK bound; all four are quarantined. Thus 84.6% is a qualification rate on the frozen panel, not a bank-wide recovery rate or a policy-benefit estimate.

The fixed-policy deployment result (Table I) does not establish an aggregate gain, safety, or equivalence. It compares tracking of raw and repaired targets; it is not a retraining effect or improved tracking of the unchanged original reference.

D. Completed allocation result: inconclusive

The earlier E4 absolute-progress arm reaches mean post-warm-up TV 0.0297, below its 0.05 gate. Its disposition is `not_tested`; it supplies no policy null. The newer confirmation retains A under its own comparator rule. R changes exposure in all three seeds (mean TV 0.08348/0.08249/0.08233), as does D (0.08391/0.08587/0.08398). All twelve training runs pass their declared gates with zero recorded invalid or censored events. Across 492 saved states, exact sampler and checkpoint replay establishes intervention integrity.

TABLE I

REPAIR QUALIFICATION AND DEPLOYMENT ANSWER DIFFERENT QUESTIONS.

Evidence / status	Result and scope
Qualification / measured	22/26 flagged candidates; 4/4 feasible controls byte-identical.
Deployment / exploratory	One policy, 26 clips, 656 paired conditions per arm: raw 0.3925, repaired 0.3915; difference −0.0010, clip-bootstrap 95% CI [−0.0086, +0.0080].

TABLE II

FINAL TRACKINGSCORE DIFFERENCES. THREE PAIRED TRAINED SEEDS; TWO-SIDED SEED T INTERVALS, DF=2.

R−U	Feasible-hard	All-panel
Seed 21	−0.03400	−0.04892
Seed 22	−0.03299	−0.01402
Seed 23	+0.02178	+0.03500
Mean	−0.01507	−0.00932
95% lower	−0.09436	−0.11405
95% upper	+0.06422	+0.09542

The frozen primary result is *inconclusive* (Table II). Neither the primary rule nor the broad non-regression guard passes. Two hard-panel seeds favor U and one favors R. The interval spans both meaningful harm and benefit; it does not establish either, nor does failure of the guard prove regression. The supplementary paired bootstrap hard interval is $[-0.04417, +0.02532]$ and cannot override this decision. Descriptive final hard R−D is -0.00675 with interval $[-0.02189, +0.00840]$; R−A is -0.00668 with interval $[-0.07243, +0.05907]$. These contrasts do not establish equivalence or progress-specific superiority.

Exploratory normalized hard AULC R−U is negative in all three seeds: -0.03978, -0.03042, -0.01676 (mean -0.02898; descriptive 95% t interval $[-0.05774, -0.00023]$). Two R seeds never attain their paired U final score on the observed grid. These data do not support an efficiency-accelerator claim; sparse checkpoints do not identify a precise crossover. The secondary interval is not a new confirmatory harm test. Figure 3 reports the complete curves.

a) Earlier controls remain separate: Table III separates prior protocol statuses. None is pooled with the completed four-arm result or substituted for a matched admission on/off test. Segmentation retained 12.5 of the 20.2 minutes in 99 flagged training clips and lost only three whole clips at zero guard. Retained duration is not demonstrated retained skill. The separate H1 comparison below estimates admission under D on a common candidate partition.

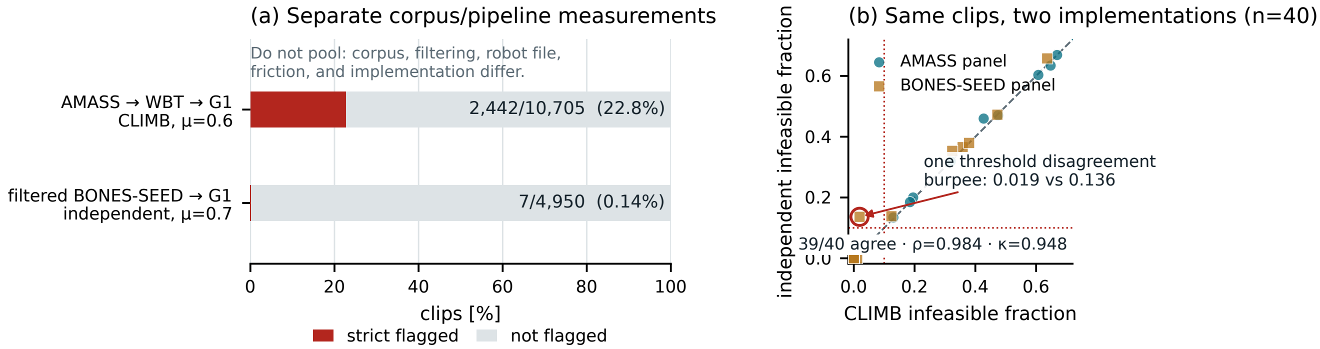

Fig. 2. Bank-scale screen and implementation agreement. (a) Bars are normalized within two separate corpus/pipeline denominators and apply the strict `infeasible_frac`> 0.10 rule: 2,442/10,705 for AMASS→WBT→G1 and 7/4,950 for filtered BONES-SEED→G1. Corpus, filtering, robot file, friction, and implementation differ, so this is not a causal retargeter comparison. (b) On a flag-enriched, deterministic 20+20 same-clip panel, strict decisions agree for 39/40 clips (Spearman $\rho = 0.9836$, Cohen's $\kappa = 0.9485$). That panel checks implementation agreement; its stratified selection is not a prevalence sample.

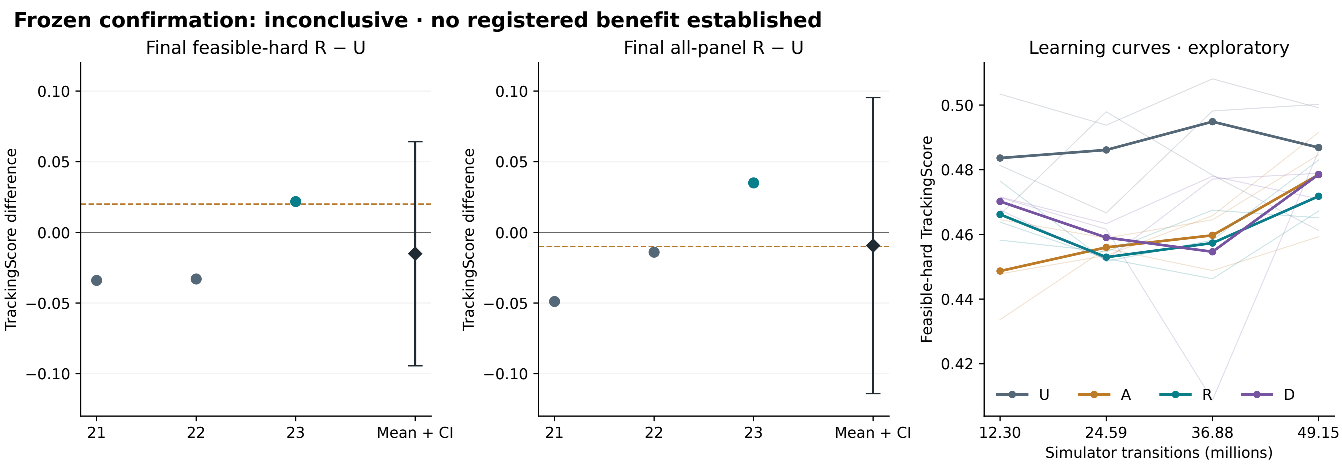

Fig. 3. Completed allocation confirmation. Seed pairs, means and two-sided 95% t intervals appear at left and center. Dashed lines are the hard point-estimate target and all-panel lower-bound margin. Thin learning curves are seeds, thick curves are means; the cost axis is simulator transitions. Curves and attainment are exploratory.

TABLE III
EARLIER CONTROLS, WITH THEIR ORIGINAL SCOPE AND STATUS.

Study / status	Bounded finding
E-HYG / sealed	Whole-clip pruning: feasible-panel survival 0.918 → 0.907; $\Delta = -0.0101$, one-sided permutation $p = 0.951$.
Soft weighting / failed gate	Hard-rejected mass 0.199; this does not implement exact admission.
E4 / not tested	Mean TV 0.0297 below 0.05; policy endpoint unopened.
Repair-all / exploratory	Deployment contrast +0.0397 misses +0.05 target and coverage gate.

E. H1: admission under D

All ten full-budget training runs and forty held-out cells pass their gates: 410 saved training states replay exactly, and 112,000 evaluation rows preserve paired initial conditions and unchanged inference tensors. Admission on has zero rejected trials; off has positive rejected exposure and respects the declared 0.0923 probability floor. Both have zero invalid or censored events.

The registered disposition is *inconclusive*. Final hard on-minus-off seed differences are $-0.0100/+0.0100/+0.0200/-0.0200/+0.0000$; mean $+0.0000$, 95% seed-t interval $[-0.0196, +0.0196]$. The interval spans zero; admission benefit, harm and equivalence remain unestablished. All-panel mean is $+0.0000$, interval $[-0.0196, +0.0196]$; this descriptive result has no non-regression pass/fail guard.

Gate-off's rejected share of positive excess allocation averages 40.0% across seeds (range 40.0–40.0%), compared with 11.50% of legal starts in its prior. This is a ratio of checkpoint-summed positive excess, not the fraction of all training samples and not a causal estimate of wasted practice. Figure 4 shows every seed. The intervention estimates admission under D; it cannot identify an admission-by-R interaction or which ingredients caused the historical E1 collapse.

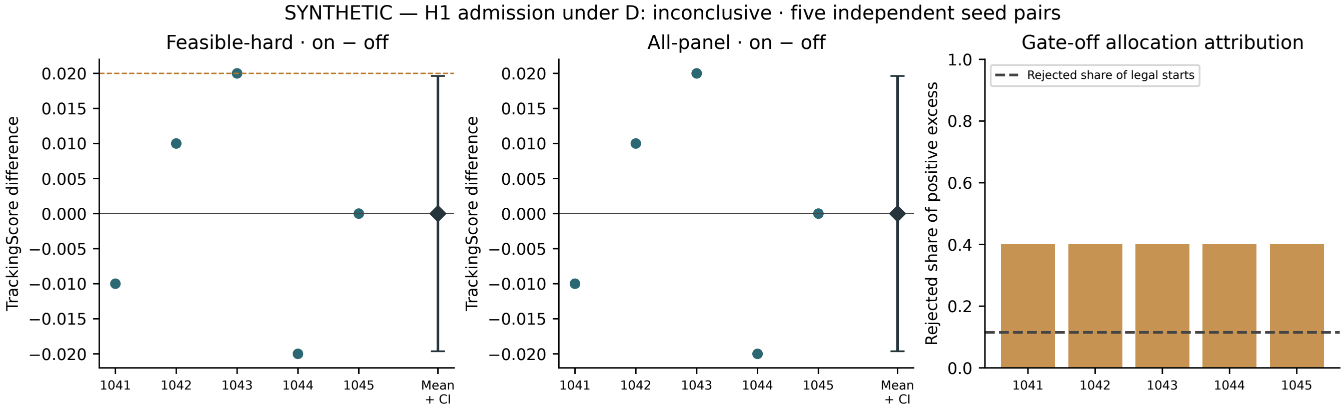

Fig. 4. H1 admission on/off under D: five paired trained seeds and their mean with two-sided 95% seed-t interval (df=4). The hard-panel dashed line is a reference, not a pass/fail target. At right, rejected share of positive excess above the prior uses the frozen post-warm-up checkpoint grid; its dashed line is the rejected legal-start prior share. Allocation attribution is distinct from policy utility.

VI. Limitations

CLIMB's admission label is model-relative. The current screen instantiates one G1 model and a flat scene with collision contacts, friction, and actuator force limits. Hardware deployment would add electrical, thermal, compliance, latency, and structural constraints and then require closed-loop validation. Terrain-bearing references likewise require their terrain in the scene; the four box-jump flags in the production bank are routed to missing context rather than declared bad motion.

Finite differences, the 6 cm contact band, and the half-weight residual threshold introduce modeling choices. Local sensitivity checks bound the two thresholds on this bank, but a new robot or retargeter must recalibrate them. The 22.8% and 0.14% rates therefore remain separate corpus–pipeline measurements. A smooth feasibility weight derived from residual slack could retain borderline cases, but would couple admissibility and allocation; it requires its own matched ablation against the current binary gate.

The allocation study has three independent seed pairs; H1 has five separate pairs. Neither the shared held-out panel nor the number of evaluation conditions increases these counts. Its observed hard-panel standard deviation 0.03192 yields a 95% t half-width 0.07929, much larger than the +0.02 point target. This is an observed precision limitation, not a retrospective power guarantee or evidence of equivalence. Thousands of evaluation conditions cannot replace independent training. The current result neither establishes benefit nor rules out a target-sized benefit; a future replication requires its own fixed sample size.

The current simulator has knee effort clamps of approximately $\pm 139\,\mathrm{N\,m}$, zero added command delay and plane terrain; physical G1 transfer is untested. Numerical parity remains unresolved in a separate physical-sensitivity development check, so its perturbation scores are not reported as evidence. A continuous adapter traverses 8.58-second training references, but comparative full-motion execution is untested. The frozen-policy instrumentation pilot does not establish a noise floor; its planned 17.6-million-transition burn-in makes it a separate study. Practice interventions, reliability-calibrated progress and hardware evaluation remain future work, not explanations established by the current allocation result.

VII. Conclusion

RPM makes persistent failure an ambiguous instruction for practice. The diagnosed campaign, bank-scale screen and cross-policy difficulty transfer motivate explicit measurement of reference admissibility. Exact temporal support then permits a controlled allocation test: relative progress changes exposure, but the completed three-seed comparison does not establish its registered tracking benefit. A qualified repair likewise does not establish a fixed-policy gain. Reference qualification, exposure and control utility therefore need separate evidence. The separate five-pair admission test under D is inconclusive; its claim is limited to that intervention, model and held-out panel.

Reproducibility and Acknowledgments

The stack uses mjlab v1.6.0 at commit `0fb8a681136b` and MuJoCo 3.11.0. Motion, sidecar, unit-table, task, checkpoint, evaluator, condition and output identities are SHA-256 bound. Licensed motion payloads cannot be redistributed; release requires reconstruction instructions and permitted aggregate metadata. OpenAI Codex and Anthropic Claude assisted with code, analysis tooling, figures and manuscript drafting in Sections I–VII; claims are tied to verified artifacts, with final author review required before submission.

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